

Sensitivity

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Reference

<https://engineeringlibrary.org/reference/thick-pressure-vessels-air-force-stress-manual>

<https://apps.dtic.mil/sti/pdfs/AD0759199.pdf>

Cylinder

Lamé equations

for Cylinder we have the following stress equations

radial tension

$$T_3(r) = A - \frac{B}{r^2}$$

circumferential tension

$$T_1(r) = A + \frac{B}{r^2}$$

axial tension (end-capped cylinder)

$$T_2(r) = A$$

whereby

$$A = \frac{P_a a^2 - P_b b^2}{b^2 - a^2}$$

$$B = \frac{a^2 b^2 (P_a - P_b)}{b^2 - a^2}$$

Assume for hydrophones internal pressure $P_a = 0$, then the two constants A and B become

$$A = -P_b \frac{b^2}{b^2 - a^2}$$

$$B = -P_b \frac{a^2 b^2}{b^2 - a^2} = A a^2$$

and the three stress equations are then

$$T_3(r) = A(1 - \frac{a^2}{r^2})$$

$$T_1(r) = A(1 + \frac{a^2}{r^2})$$

$$T_2(r) = A$$

with A as defined above

Electric field

The electric field in radial direction $E_3(r)$ due to external pressure becomes with $g_{32} = g_{g1}$

$$E_3(r) = -g_{31}(T_1(r) + T_2(r)) - g_{33}T_3(r)$$

and the measured voltage V

$$V = \int_a^b E_3(r) dr$$

Note:

$$\int_a^b dr = b - a$$

$$\int_a^b \frac{1}{r^2} dr = -(\frac{1}{b} - \frac{1}{a}) = \frac{b-a}{ab}$$

that is

$$\int_a^b (1 \pm \frac{a^2}{r^2}) dr = (b-a) \pm \frac{a(b-a)}{b} = (b-a)(1 \pm \frac{a}{b})$$

Generated voltage

$$V = - \int_a^b (g_{31}(T_1(r) + T_2(r)) + g_{33}T_3(r)) dr$$

$$V = -A(b-a)(g_{31}(2 + \frac{a}{b}) + g_{33}(1 - \frac{a}{b}))$$

$$V = -P_b b \frac{b}{b+a} (g_{31}(2 + \frac{a}{b}) + g_{33}(1 - \frac{a}{b}))$$

let $t = \frac{a}{b}$ then

$$V = -P_b b \frac{1}{1+t} (g_{31}(2+t) + g_{33}(1-t))$$

Sphere

$$T_3(r) = P_b \frac{b^3}{a^3 - b^3} (1 - \frac{a^3}{r^3})$$

$$T_1(r) = T_2(r) = P_b \frac{b^3}{a^3 - b^3} (1 + \frac{a^3}{2r^3})$$

With

$$\int_a^b \frac{1}{r^3} dr = -\frac{1}{2} (\frac{1}{b^2} - \frac{1}{a^2}) = \frac{1}{2} \frac{b^2 - a^2}{a^2 b^2} = \frac{1}{2} (\frac{b+a}{ab}) (\frac{b-a}{ab})$$

and therefore

$$\int_a^b (1 - \frac{a^3}{r^3}) dr = (b-a) - \frac{a^3}{2} \frac{b+a}{ab} \frac{b-a}{ab} = (b-a) (1 - \frac{a}{2b} \frac{b+a}{b}) = \frac{(b-a)}{2} (2 - \frac{a}{b} - \frac{a^2}{b^2})$$

$$\int_a^b (1 + \frac{1}{2} \frac{a^3}{r^3}) dr = (b-a) + \frac{1}{2} \frac{a^3}{2} \frac{b+a}{ab} \frac{b-a}{ab} = (b-a) (1 + \frac{1}{2} \frac{a}{2b} \frac{b+a}{b}) = \frac{(b-a)}{4} (4 + \frac{a}{b} + \frac{a^2}{b^2})$$

one gets

$$V = -P_b \frac{b^3}{b^3 - a^3} \frac{(b-a)}{2} (g_{31}(4 + \frac{a}{b} + \frac{a^2}{b^2}) + g_{33}(2 - \frac{a}{b} - \frac{a^2}{b^2}))$$

and finally

$$V = -P_b \frac{b}{2} \frac{b^2}{b^2 + ab + a^2} (g_{31}(4 + \frac{a}{b} + \frac{a^2}{b^2}) + g_{33}(2 - \frac{a}{b} - \frac{a^2}{b^2}))$$

which again is equivalent to

$$V = -P_b \frac{b}{2} \frac{1}{1+t+t^2} (g_{31}(4+t+t^2) + g_{33}(2-t-t^2))$$

Bubble resonance under pressure

Boyle's equation

$$P(z)V(z) = P_0 V_0 = \text{const}$$

Bubble volume

$$V(z) = \frac{4}{3} \pi a^3$$

$$V(z) = V_0 \frac{P_0}{P(z)} = \frac{V_0}{(1 + 0.1z)}$$

$$a(z) = \frac{a_0}{(1 + 0.1z)^{1/3}}$$

Minnaert resonance

$$f = \frac{1}{2\pi a} \left(\frac{3\gamma P_A}{\rho} \right)^{1/2}$$

$$f(z) = \frac{1}{2\pi a} \left(\frac{3\gamma P_0}{\rho} \right)^{1/2} (1 + 0.1z)^{1/2}$$

$$P_0 = 100 \text{ kPa}$$

$$\rho = 1000 \text{ kg/m}^3$$

$$\gamma = 1.4$$

$$f(z) \times a = 3.26(1 + 0.1z)^{1/2} \text{ m/s}$$

Clay and Medwin (from Cato)

$$f = 3.25 \frac{(1 + 0.1z)^{1/2}}{a}$$

test

$$f = 3.26 \frac{(1 + 0.1z)^{1/2}}{a_0} (1 + 0.1z)^{1/3}$$

$$f = 3.26 \frac{(1 + 0.1z)^{5/6}}{a_0}$$